

Now add some virtual objects like a capsule to the scene by typing

```
• let capsuleNode = SCNNode(geometry: SCNCapsule(capRadius: 0.05, height: 0.3)
```

You can position the object by dragging it along the axis or by just typing in a separate command to shift the nodes:

```
• capsuleNode.position = SCNVector3(0.3, 0.3, 0.5)
• sceneView.scene.rootNode.addChildNode(capsuleNode)
```

Now run the app and make some changes to the system simultaneously to see how the environment reacts.

```
• capsuleNode.geometry?.firstMaterial?.diffuse.contents = UIColor.Green
•
• capsuleNode.eulerAngles = SCNVector3(0,0,Double.pi/2)
```

The above commands will change the colour of the capsule to blue and change the node's angular position as per the dimensions.

```
• func renderer(SCNSceneRenderer, didAdd: SCNNode, for: ARAnchor)
• func renderer(_ renderer: SCNSceneRenderer, didUpdate node: SCNNode, for anchor: ARAnchor)
• func renderer(SCNSceneRenderer, didRemove: SCNNode, for: ARAnchor)
• config.planeDetection = .horizontal
• method, add the below line.
• sceneView.delegate = self
```

The func renderer commands help in correlating the virtual positions of the object to its actual position along with rendering parameters like position and node visibilities.

## AR Buildup Topology

Implement the Ad SCNNode delegate method to configure and add elements for child elements.